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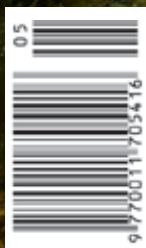
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# America's Battle of the Atlantic

New mapping technology and a small manned submersible allow a team of scientists to 3D map a moment in history



USCG cutter dropping depth charges

by Joseph Frey

On July 15, 1942, Kapitänleutnant Hans-Dieter Heinicke, commander of German submarine U-576, would make a decision that would seal his fate and that of his 44 crew members and four Allied sailors off North Carolina's Outer Banks. In the process he would leave an amazing underwater archaeological record of a Second World War convoy battlefield.

**I**n August 2016, racing against tropical storms and hurricanes, marine archaeologists from the National Oceanic and Atmospheric Administration (NOAA) and the Bureau of Ocean Energy Management (BOEM) were amongst the first to lay eyes on U-576 and its target, the merchant ship, *Bluefields*, since they were sunk during that July naval battle 74 years ago.

I was invited to participate on this expedition by Dr. James Delgado, NOAA's Director of Maritime Heritage and I was extremely excited to be diving on U-576 in one of the expedition's two Triton 1000-2 submersibles. We're 56 km to the east of Cape Hatteras in just over 213m in the expedition's support ship, the R/V *Baseline Explorer*.

## War casualties

The story of the Battle of the Atlantic Expedition (named by Joe Hoyt, the Senior Maritime Archaeologist for NOAA's Monitor National Marine Sanctuary) really begins in 1941 when U-576, a Type VIIC Unterseeboot (U-boat) is launched under the command of Lieutenant-Commander Heinicke. Initially based in Kirkenes, Norway, U-576 conducts two war patrols without contacting merchant shipping. Relocated to Saint-Nazaire in France sees things change when, in January 1942, Heinicke and his crew are sent to attack shipping along the Canadian and US coasts. U-576 sinks its first freighter, the *Empire Spring*, off Sable Island, Nova Scotia then, returning to North America for its fourth patrol during April 1942, U-576 sinks two

more freighters, the *Pipestone County* and the *Taborfell*.

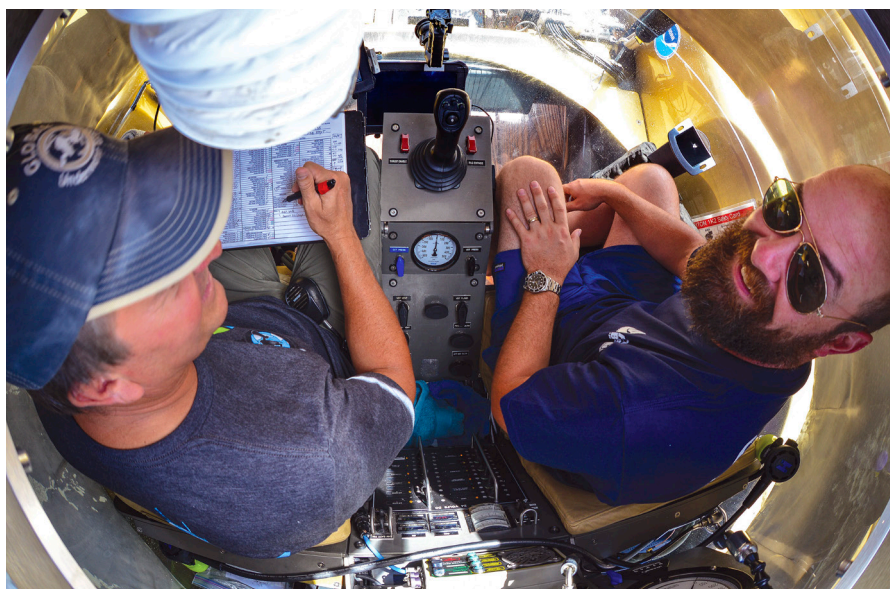
It's U-576's fifth war patrol, during July 1942 along America's east coast that marks the beginning of the end for Heinicke and his crew. After repairing engine problems and a damaged ballast tank 160 km out at sea, Heinicke then returns U-576 to the Outer Banks and actively hunts Convoy KS-520 with its 19 armed merchant ships and five US Navy and US Coast Guard escorts off Cape Hatteras.

Why U-576 on its own would attack a well-defended convoy is open to speculation, but U-576 fires four torpedoes sinking the *Bluefields* and damaging two other freighters. Surfacing possibly inadvertently, U-576 finds itself in the middle of



the convoy where within minutes it is sunk by a combination of deck gunfire from the freighter *Unicoi* and depth charges dropped by two US Navy Kingfisher aircraft. Both *Bluefields* and U-576 sink to their watery graves in minutes, and within 224m of each other.

“The U-576 and *Bluefields* sites are unique in their proximity to each other, representing a convoy battle,” says Joe Hoyt. “The Battle of the Atlantic project is motivated by recognition that these resources are unique and under-represented in popular history.”



Submersible pilot Robert Carmichael and NOAA archaeologist Joe Hoyt

### The human element

With this in mind, there is a move to place U-576, an official German war grave, and the *Bluefields* under the protection of the Monitor National Marine Sanctuary through a process involving public consultations with the fishing and recreational diving communities and others.

NOAA's search for U-576 and the *Bluefields* began eight years ago and started to bear results during 2013 when a target suspected to be *Bluefields* was spotted using side scan sonar. Both *Bluefields* and the

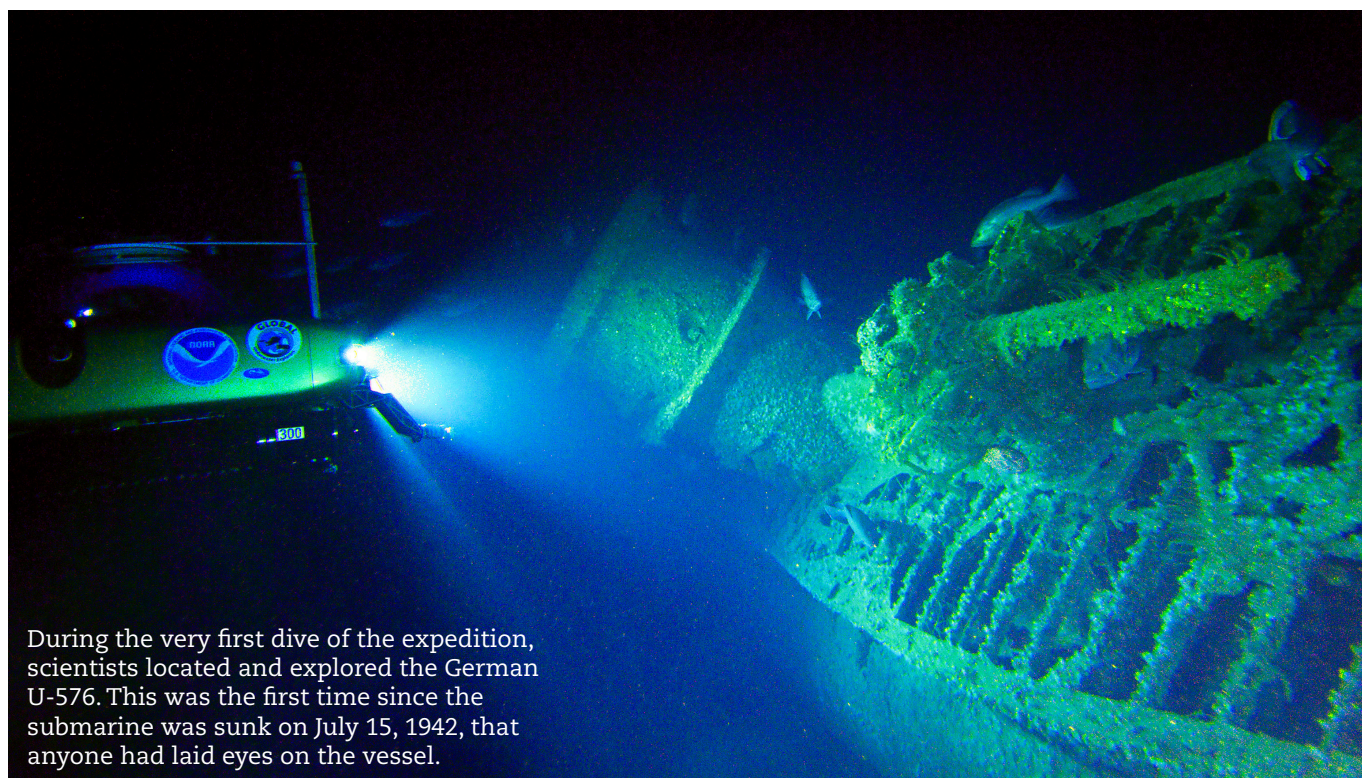
surrounding area that contained U-576 were imaged in a follow-up survey using ship-based and AUV multi beam technology in 2014. Now, two years later, it was to be the first time in 74 years that human eyes would view these two casualties of the Second World War.

We would use two two-person Triton submersibles to view U-576 and the *Bluefields* as well as to collect scientific data. While much of the data could be collected from alternative platforms such as AUVs and ROVs, submersibles give

archaeologists a true first hand understanding of the site.

“A major element of our work is geared towards preservation, which requires public engagement,” Hoyt says. “Getting people to join in for the journey is critical to get them to care about the work.”

Mounted on the Triton submersibles are ULS-500 spread beam laser scanners from Canadian company 2G Robotics that are combined with a positioning solution from Britain's Sonardyne. The cameras



During the very first dive of the expedition, scientists located and explored the German U-576. This was the first time since the submarine was sunk on July 15, 1942, that anyone had laid eyes on the vessel.



are housed in depth-rated housings made by another Canadian firm, SubC Imaging. Sonardyne's advanced acoustic-inertial navigation technology, combined with 2G Robotics' ULS-500 laser profiling sensors allow highly time efficient 3D mapping to millimetre or centimetre levels of accuracy and resolution in any water depth. The technology is brand new, and the resulting mapping capability previously inconceivable. NOAA, in conjunction with 2G Robotics realized that laser-acoustic-inertial mobile mapping would be applicable to underwater archeology. This is believed to be the world's first application of it from small manned submersibles.

Dane Mikael Larsen is Sonardyne's principal engineer for the firm's acoustic inertial systems. A PhD, with 20 years' experience in high accuracy underwater navigation, Larsen took part in one of the first dives to U-576 in order to optimise the operation of acoustic-inertial and laser systems for the mapping of the two wrecks.

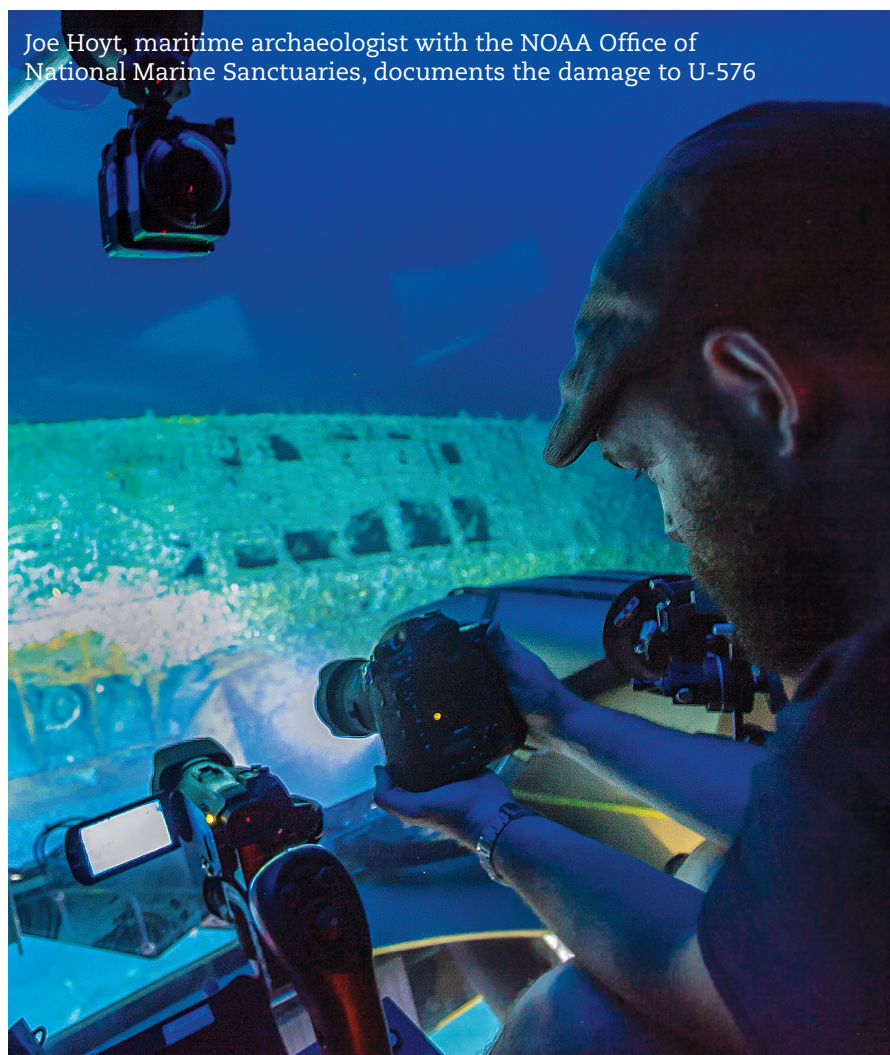
Diving on the U-576 was a completely overwhelming experience for Larsen when, out of the darkness, the U-boat appeared. Seeing the fine and perfectly preserved details of the U-boat first hand was somehow a lot more powerful than seeing it through the video cameras of an ROV. The U-boat appeared undamaged, apart from 74 years of marine growth. The personal high point of the dive for Larsen was towards the end when the pilot halted the submarine on the sandy sea-floor just next to the U-boat conning tower. "We switched off all lights and instrumentation and sat in total silence and darkness, apart from the weak blue ambient light penetrating 225 metres from the surface. As if by magic, the entire hull from bow to stern slowly stood out as our eyes adjusted to the darkness."

"The extreme tranquility of the environment was in eerie contrast to the purpose of the U-boat and the mental picture of the violent and



Hoffman getting ready to dive

...As if by magic, the entire hull from bow to stern slowly stood out as our eyes adjusted to the darkness...



Joe Hoyt, maritime archaeologist with the NOAA Office of National Marine Sanctuaries, documents the damage to U-576



chaotic battle that took place at the surface years before, but it somehow appeared perfectly fitting and respectful considering the site is now the final resting place of so many US and German sailors who sacrificed their lives. Even the numerous groupers patrolling the U-576 and *Bluefields* appeared to show their respect by their slow and majestic movement.”

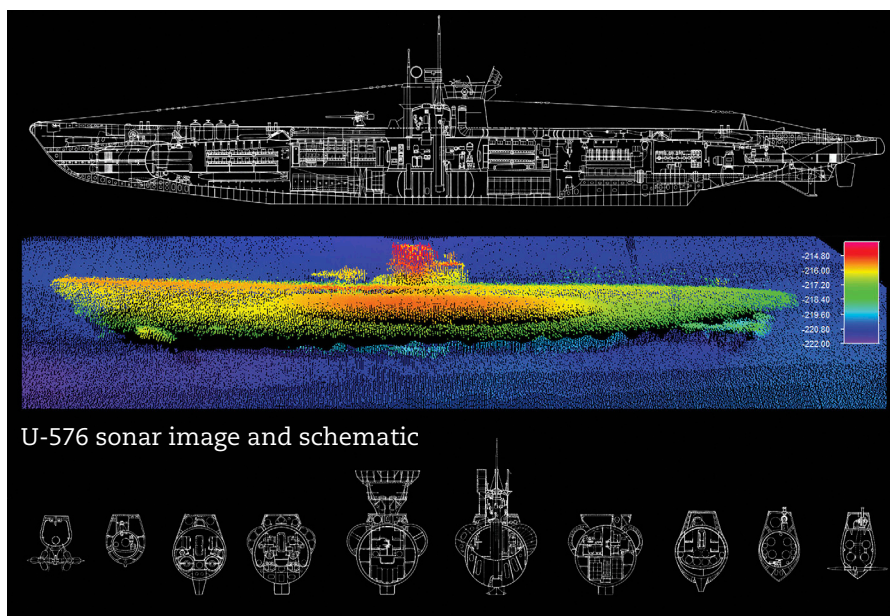
### Photogrammetric modelling

It's the morning of August 28 and William Hoffman, a BOEM maritime archaeologist, is getting ready for the day's second submersible dive. I'm scheduled too, to dive that afternoon. The weather forecast is good.

Hoffman has been a certified scuba diver since 1999 and currently serves as the Atlantic Regional Diving Officer for BOEM's Scientific Diving Program. While I've had previous experience diving in the Royal Canadian Navy's former SDL-1 submersible, this is Hoffman's first submersible dive. It's a great opportunity to experience firsthand the application of manned submersibles equipped with laser scanning payloads and other equipment for documenting archaeological sites.

During Hoffman's dive with submersible pilot Randy Holt, they investigate U-576 and take detailed and overlapping photographs to create later a photogrammetric model of the site. The photogrammetry complements the laser scanning conducted at the site and together they allow for 3D visualization and modeling of U-576 to an incredible level of accuracy and detail.

They descend to the seafloor and slowly begin to transit to the site guided by distance and range calculations from R/V *Baseline Explorer's* Surface Officer Shane Ziegler. On the descent, Hoffman was pre-occupied with making sure the camera and lighting systems are operational and configured properly, and it wasn't until they reached the seafloor and Holt cut the power to the camera strobes and sub lights to



U-576 sonar image and schematic

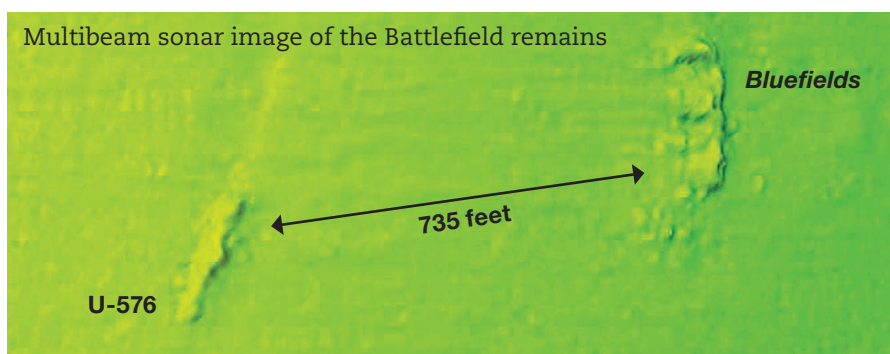


Photo: Ed Caram, NOAA and SRI International

conserve power that Hoffman has a moment to take it all in. It's quiet and strangely peaceful and he is amazed by the amount of ambient light still penetrating to the 225m depth they are sitting at. About 9.5 m away from U-576 they make out the shape of the hull looming out of the darkness. The seafloor is partially scoured around the site, and the bow and stern of the hull partially suspended above the

seafloor, appearing to float, ghost-like.

“We did observe an area of damage to the outer hull fairing on the starboard side of the U-576 but it is not yet clear whether that damage occurred during the sinking or is post-depositional,” Hoffman reports. “During photogrammetry we were also able to observe the keel and



U-576 conning tower anti-aircraft gun mount



Conning tower of U-576



Photo courtesy: Robert Carmichael, Project Baseline

ballast tanks up close and we could see no obvious damage.

“Overall, the hull appears to be exceptionally intact. The watertight hatch in the conning tower is visible and is clearly closed and sealed, and there are no visible clues to indicate that the crew made an attempt to escape. Now that the fieldwork is completed, the work of post-processing the data, completing the laser scanning model, and stitching together the photogrammetric model is taking place. There will be many hours spent poring over the data to see what else can be learned.”

Excitement builds as I watch Hoffman’s submersible hauled out of the Atlantic Ocean and onto the stern of *Baseline Explorer* but before Hoffman can even set foot onto the ship’s deck we’re given instructions that all personnel not needed to sail the *Baseline Explorer* must depart on two smaller, faster vessels to Ocracoke Island - Tropical Depression Eight is rapidly forming to the south of us. Tropical Storm Hermine becomes Hurricane Hermine and moves north to the Outer Banks.

As it turns out, August 28 marks the last day of the expedition at sea. Fortunately, the quality of data returned from the survey site by 2G Robotics spread beam laser triangulation system exceeds expectations; combined with the optimized navigation data from Sonardyne they will generate 3D maps to add to the process to have U-576 and *Bluefields* included in the Monitor National Marine Sanctuary as a memorial to the American Theatre of the Second World War.

(Reprinted courtesy of *DIVER* magazine Canada)



U-576 in rough seas

Stern cabin of *Bluefields*



Photo: John McCord, UNC Coastal Studies Institute, NOAA



NOAA archaeologist John Bright, pilot, Randy Holt, and swimmer Austin Adams return to *Baseline Explorer*